

Catalytic Aerobic Oxidation of *p*-Methoxybenzyl (PMB) Ethers to Aldehydes or Ketones

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An efficient and highly selective aerobic oxidation of *p*-methoxybenzyl ethers to aldehydes or ketones with Fe(NO₃)₃·9H₂O and TEMPO as catalysts at 25 °C has been developed. This method is compatible with versatile functional groups: adamantyl, ether, alkynyl, and alkenyl, etc. Due to the mild nature of the reaction conditions, even the highly sensitive optically active aldehydes with an easily racemized α-chiral

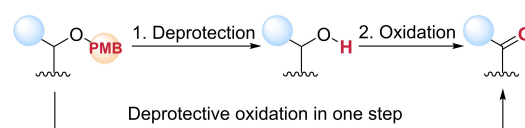
center could be formed from optically active PMB ethers without racemization. Other protecting groups of hydroxyl group remained intact during the current aerobic oxidation. The catalytic protocol has been successfully applied to synthesis of (*R*)-*tert*-butyldimethylsilyl hept-1-en-6-yn-4-yl ether, which is the key intermediate for the total synthesis of natural product, (–)-6-*O*-methylcitricisocoumarin.

Introduction

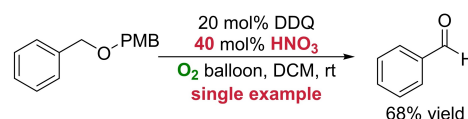
Hydroxyl groups are a class of common, versatile, and reactive functional groups in organic chemistry. In organic synthesis, the hydroxyl groups are usually protected when a targeted reaction is to be carried out selectively with another functionality. *p*-Methoxybenzyl (PMB) ethers are among the most frequently used protecting groups in organic synthesis.^[1] In multi-step synthesis, the alcohols would be protected as PMB ethers, which were deprotected usually with DDQ (2,3-dichloro-5,6-dicyano-1,4-benzoquinone),^[2a–k] Li/NH₃,^[2l] or TFA (trifluoroacetic acid)^[2m] first to release the hydroxyl group. Subsequent oxidation with stoichiometric amounts of oxidants such as TPAP/NMO (tetrapropylammonium perruthenate/*N*-methylmorpholine oxide),^[2a] PCC (pyridinium chlorochromate),^[2b,c] dimethyl sulfoxide,^[2d,e] DMP (Dess–Martin periodinane),^[2f–i] BAIB ((diacetoxyiodo)benzene),^[2k] and IBX (2-iodylbenzoic acid)^[2m] would afford aldehydes or ketones (Scheme 1a). Such synthetic operation lacks the step economy and is environmentally unfriendly. Molecular oxygen, an important component of air, is eco-friendly and inexpensive. Thus, catalytic aerobic oxidation of PMB ethers to carbonyl compounds would be an ideal solution.

Recently, an aerobic oxidative reaction of PMB ethers to carboxylic acids has been established,^[3] however, the aerobic oxidative reaction of PMB ethers to aldehydes or ketones is more challenging due to the stronger electron-withdrawing ability and the reactivity of aldehydes or ketones. In fact, there

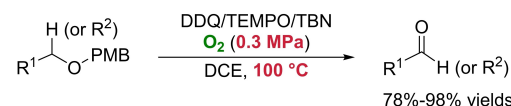
(a) Multi-step vs. One step



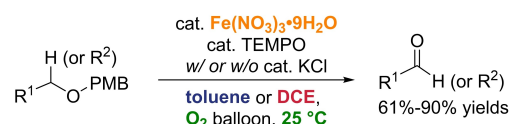
(b) DDQ/HNO₃-cocatalyzed aerobic oxidation of PMB ether to benzaldehyde



(c) DDQ/TEMPO/TBN-cocatalyzed aerobic oxidation of PMB ethers to aldehydes or ketones



(d) This work: Iron/TEMPO-cocatalyzed aerobic oxidation of PMB ethers to aldehydes or ketones



Scheme 1. The traditional strategies for transforming PMB ethers to aldehydes or ketones and catalytic aerobic oxidations of PMB ethers to aldehydes or ketones.

are only sporadic reports on such a transformation: Arseniyadis and Jean reported aerobic oxidation of single example of benzyl *p*-methoxybenzyl ether to benzaldehyde with 20 mol% of DDQ and 40 mol% of HNO₃ (Scheme 1b).^[4a] In 2015, Hu and Shen successfully applied a catalytic amount each of DDQ/TEMPO/TBN for aerobic of PMB ethers to aldehydes or ketones at 100 °C with 0.3 MPa of oxygen (Scheme 1c).^[4b] As mentioned, the challenge is the high reactivity of or caused by the formed ketones or aldehydes: For example, in 2017, Moriyama and his coworkers reported that under the catalysis of sodium nitrite and 4-nitrobenzenesulfonamide, 4-chloro- α -methylbenzyl PMB

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ether underwent deprotection, aerobic oxidation, and subsequent bromination to afford 4-chloro- α -bromophenylethanone at 50 °C with the help of 5.0 equiv. of HBr.^[4c] In 2021, Khan and his coworkers reported DDQ-catalyzed deprotective oxidation and subsequent dehydrogenation of (5,6,8-trimethoxy-4-phenyl-1,2,3,4-tetrahydronaphthalen-2-yl)methyl PMB ether in 48% yield.^[4d] Thus, the direct efficient conversion of PMB ethers to aldehydes or ketones under ambient conditions is of high current interest. Herein, we report a Fe(NO₃)₃·9H₂O/TEMPO-cocatalyzed direct deprotective aerobic oxidation of PMB ethers to aldehydes or ketones at 25 °C with 1 atm of oxygen (Scheme 1d).

Results and Discussion

When using Fe(NO₃)₃·9H₂O and TEMPO as catalysts,^[5] the reaction of PMB ether of dodecanol **1a** in toluene was very slow (Table 1, entry 1). With 6 mol% KCl as the additive, the reaction rate was slightly increased (Table 1, entry 1 vs. entry 2). Further increasing the amount of KCl to 12 mol% led to 74% NMR yield of aldehyde **2a** and 18% NMR yield of carboxylic acid **3a** (Table 1, entry 3). Surprisingly, upon increasing KCl to 18 mol%, 71% NMR yield of dodecanal **2a** was obtained with an exclusive selectivity for the formation of aldehyde **2a** over carboxylic acid **3a**, although 9% recovery of PMB ether **1a**

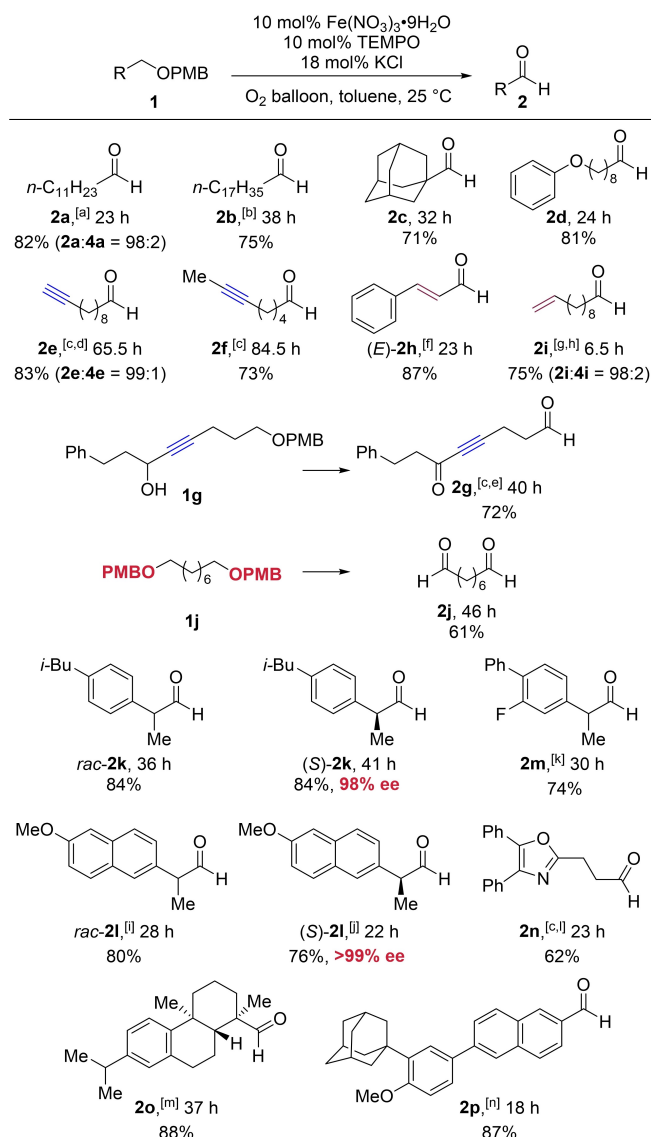
remained. In addition, nitrite **1a''** was also formed (Table 1, entry 4). There is a striking solvent effect since the reaction in DCM or MeCN afforded carboxylic acid **3a** as the major product (Table 1, entries 5 and 6). When MTBE (methyl *tert*-butyl ether) was applied as the solvent, the selectivity is poor forming a mixture of 30% NMR yield of aldehyde **2a**, 16% NMR yield of carboxylic acid **3a**, 22% of alcohol **1a'**, and 23% recovery of **1a** (Table 1, entry 7). The reaction with Cu(NO₃)₂·3H₂O replacing Fe(NO₃)₃·9H₂O was much slower (Table 1, entry 8). In the absence of Fe(NO₃)₃·9H₂O or TEMPO, no reaction occurred (Table 1, entries 9 and 10). We also found that pre-stirring of the catalysts was beneficial for the reaction (Table 1, entry 4 vs. entry 11). If the catalysts had not been pre-stirred before adding PMB ether, 66% NMR yield of aldehyde **2a** and 13% recovery of **1a** were detected within the same reaction time (Table 1, entry 11).

Subsequently, we examined the scope of this Fe(NO₃)₃·9H₂O/TEMPO-cocatalyzed aerobic oxidative reaction. As shown in Scheme 2, PMB ether with alkyl chain length of 12 carbon atoms gave dodecanal **2a** in 82% yield. When the length of alkyl chain was increased to 18, the reaction may also proceed smoothly to afford octadecanal **2b** in 75% yield. The substrate with a highly sterically hindered group such as adamantyl also worked affording aldehyde **2c**. Interestingly, phenyl ether group in **1d** was compatible to give 9-phenoxy-nonanal **2d** in 81% yield. In addition, substrates with terminal/

Table 1. Optimization of the reaction conditions.^[a]

Entry	X (mol %)	solvent	Recovery of 1a (%)	NMR yield (%)					
				2a	3a	1a'	1a''	4a	5a
1	0	Toluene	84	10	0	0	0	0	16
2	6	Toluene	76	18	0	0	0	0	24
3	12	Toluene	2	74	18	1	0	2	92
4	18	Toluene	9	71	0	7	4	2	85
5	18	DCM	0	28	66	0	0	3	96
6	18	MeCN	0	16	74	0	0	3	98
7	18	MTBE	23	30	16	22	8	2	78
8 ^[b]	18	Toluene	92	8	0	0	0	0	8
9 ^[c]	18	Toluene	100	0	0	0	0	0	0
10 ^[d]	18	Toluene	100	0	0	0	0	0	0
11 ^[e]	18	Toluene	13	66	0	9	5	2	84

[a] Reaction conditions: To a 25 mL flask were added KCl, TEMPO, and Fe(NO₃)₃·9H₂O sequentially. After adding 1.0 mL of solvent, the resulting mixture was stirred at rt for 5 min, then **1a** (0.5 mmol) and 1.0 mL of solvent were added. After the installation of an O₂ balloon, the resulting mixture was stirred at 25 °C for 20 h. The recovery and NMR yield were determined by ¹H NMR analysis of the crude product. [b] 10 mol% Cu(NO₃)₂·3H₂O was used instead of Fe(NO₃)₃·9H₂O. [c] The reaction was conducted in the absence of Fe(NO₃)₃·9H₂O. [d] The reaction was conducted in the absence of TEMPO. [e] The reaction was conducted without pre-stirring the catalysts.

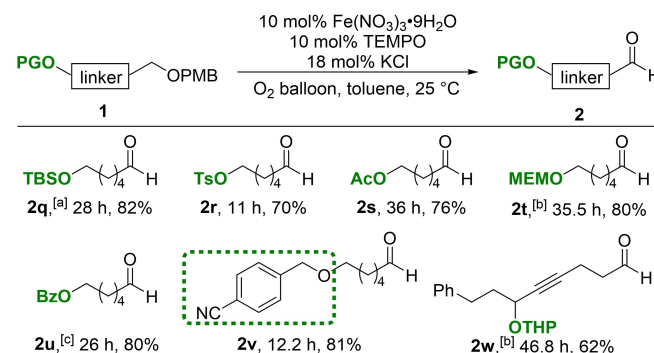


Scheme 2. Scope study. Reaction conditions: To a 25 mL flask were added 18 mol% KCl, 10 mol% TEMPO, and 10 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ sequentially. After adding 2.0 mL of toluene, the resulting mixture was stirred at rt for 5 min, then 1 (1.0 mmol) and 2.0 mL of toluene were added. After the installation of an O_2 balloon, the resulting mixture was stirred at 25 °C. Isolated yields were reported. The recovery of 1 and NMR yields of byproducts were determined by ^1H NMR analysis of the crude product. [a] 8% recovery of PMB ether 1a and 6% NMR yield of alcohol 1a', 5% NMR yield of nitrite 1a'', and 2% NMR yield of ester 4a were determined by ^1H NMR analysis of the crude product. [b] 8% recovery of PMB ether 1b was determined by ^1H NMR analysis of the crude product. [c] 10 mol% 4-OH-TEMPO and 18 mol% NaCl were used instead of 10 mol% TEMPO and 18 mol% KCl. [d] 1% yield of ester 4e was determined by ^1H NMR analysis of the crude product. [e] 3% yield of ester 4g was determined by ^1H NMR analysis of the crude product. [f] 10 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 10 mol% TEMPO, 15 mol% KCl, and 2.0 mL of DCE were used. [g] 10 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, 10 mol% TEMPO, 10 mol% KCl, and 2.0 mL of DCE were used. [h] 3% yield of ester 4i was determined by ^1H NMR analysis of the crude product. [i] 5% yield of alcohol rac-1l' and 2% yield of acid rac-3l were determined by ^1H NMR analysis of the crude product. [j] 10% yield of alcohol (S)-1l' and 2% yield of acid (S)-3l were determined by ^1H NMR analysis of the crude product. [k] 9% yield of alcohol 1m' and 3% yield of acid 3m were determined by ^1H NMR analysis of the crude product. [l] 5% recovery of 1n and 9% yield of alcohol 1n' were determined by ^1H NMR analysis of the crude product. [m] The reaction was conducted at 35 °C. [n] 0.5 mmol of 1p was used and toluene/DCM (1:2 v/v) were used as solvent.

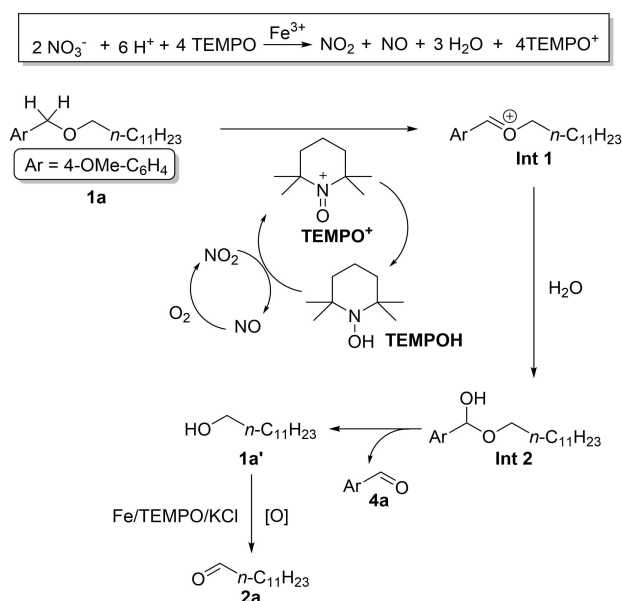
non-terminal C–C triple bond also proceeded well to afford targeted aldehydes **2e–2g** in decent yields. PMB ether **1g** containing a secondary hydroxyl group was successfully converted to the corresponding aldehyde **2g** with the secondary alcohol unit also being oxidized to ketone. Non-terminal/terminal C–C double bond, were also compatible to give corresponding aldehydes (*E*)-**2h** and **2i**. Bis-PMB ether of 1,8-octanediol **1j** afforded dialdehyde **2j**. Furthermore, the substrates derived from pharmaceuticals and natural products including ibuprofen, naproxen, flurbiprofen, oxaprozin, dehydroabietic acid, and adapalene also run well to give corresponding aldehydes. Notably, the optically active substrates with a sensitive α -chiral center also worked without racemization under standard conditions [(*S*)-**2k** and (*S*)-**2l**] due to the mild nature of the reaction conditions.

We further scrutinized the diol substrate scope of PMB ethers with different protecting groups of the remaining hydroxyl group to test the selectivity of this deprotective oxidation. As shown in Scheme 3, common protecting groups of hydroxyl group, such as OTBS, OTs, OAc, OBz, and even the acid-sensitive OMEM group were all compatible (**2q–2u**). When one of the two hydroxyl groups in 1,6-hexanediol was protected as PMB ether and the other was protected as *p*-cyanobenzyl ether (**1v**), the PMB ether was highly selectively removed affording aldehyde **2v** with the *p*-cyanobenzyl ether remained untouched. When the secondary hydroxyl group in **1w** was protected as THP ether, THP group could also be remained.

Next, the PMB ethers derived from secondary aliphatic alcohols were investigated. PMB ether **1y** was recovered in 22% yield with 44% yield of **2y** under the previously optimized conditions after 19.5 h (Table 2, entry 1). When using DCE as solvent, the reaction of **1y** could proceed smoothly to afford ketone **2y** in 95% NMR yield (Table 2, entry 2). Control experiment showed that the reaction worked better even without KCl



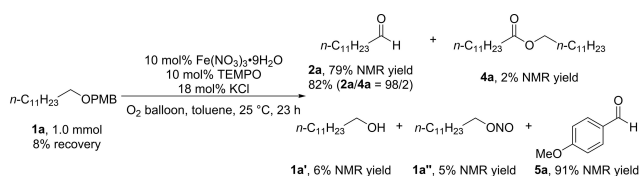
Scheme 3. Selective deprotective oxidation of PMB ethers. Reaction conditions: To a 25 mL flask were added 18 mol% KCl, 10 mol% TEMPO, and 10 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ sequentially. After adding 2.0 mL of toluene, the resulting mixture was stirred at rt for 5 min, then 1 (1.0 mmol) and 2.0 mL of toluene were added. After the installation of an O_2 balloon, the resulting mixture was stirred at 25 °C. All yields were isolated yields. The recovery of 1 and NMR yields of byproducts were determined by ^1H NMR analysis of the crude product. [a] 3% recovery of 1q and 2% yield of ester 4q were determined by ^1H NMR analysis of the crude product. [b] 10 mol% 4-OH-TEMPO and 18 mol% NaCl were used instead of 10 mol% TEMPO and 18 mol% KCl. [c] 3% recovery of 1u and 2% yield of ester 4u were determined by ^1H NMR analysis of the crude product.



Scheme 6. Proposed Mechanism.

proceeded under very mild reaction conditions (25 °C) ensuring that no racemization occurred with the reaction of optically active PMB ethers. Moreover, the key intermediate, (*R*)-*tert*-butyldimethylsilyl hept-1-en-6-yn-4-yl ether (*R*)-12, for the formal synthesis of natural product (–)-6-*O*-methylcitreisocoumarin 13 has been synthesized with this reaction as the key step. Further studies including the effect of different salts are being actively pursued in our laboratory.

Experimental Section



Caution: The flash point of toluene is +4 °C. Please strictly prohibit open flames when conducting experiments.

To a 25 mL flask were added KCl (13.3 mg, 0.18 mmol), TEMPO (15.9 mg, 0.1 mmol), and Fe(NO₃)₃·9H₂O (41.5 mg, 0.1 mmol) sequentially. After the addition of toluene (2.0 mL), the resulting mixture was stirred at rt for 5 min, then 1a (306.7 mg, 1.0 mmol) and toluene (2.0 mL) were added. After the installation of an O₂ balloon, the resulting mixture was stirred at 25 °C for 23 h until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate = 5/1), filtrated through a short column of silica gel eluted with ethyl ether (120 mL). After evaporation, 8% recovery of 1a, 79% NMR yield of 2a, 6% NMR yield of 1a', 5% NMR yield of 1a'', 2% NMR yield of 4a, and 91% NMR yield of 5a were determined by ¹H NMR analysis of the crude product with CH₂Br₂ (70.0 μL) as the internal standard. After evaporation, the residue was purified by chromatography on silica gel to afford the mixture of 2a and 4a (150.9 mg, 82%, 2a/4a = 98/2) [eluent: petroleum ether/ethyl ether = 100/1 (~400 mL)] as colorless oil: ¹H NMR

(400 MHz, CDCl₃) δ 9.77 (t, *J* = 1.8 Hz, 1 H, CHO), 2.42 (td, *J*₁ = 7.3 Hz, *J*₂ = 1.7 Hz, 2 H, CH₂), 1.71–1.55 (m, 2 H, CH₂), 1.40–1.18 (m, 16 H, 8 x CH₂), 0.88 (t, *J* = 6.8 Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃) δ 202.9, 43.9, 31.9, 29.5, 29.4, 29.32, 29.28, 29.1, 22.6, 22.0, 14.1; MS (70 eV, EI) *m/z* (%): 184 (M⁺, 2.64), 57 (100); IR (neat): ν = 2924, 2853, 2710, 1727, 1463, 1411, 1389 cm⁻¹.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: PMB ethers · Aldehydes · Ketones · Aerobic oxidation · Iron catalysis

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